

Computer Systems, B1-2 2026-27

Introduction

David Marchant

DIKU, August 31st, 2026

Course structure

Course Content

Representing information as bits

Everything is bits

Bit-level manipulation

Integers

- David on the Discord, david.marchant@di.ku.dk, 772-01-0-s06
- Mit dankse er ikke godt
- I will make an appearance at exercise classes and assignment cafes
- I run Linux, not Mac, not Windows
- Questions and comments are always welcome

Lecturers



Michael Kirkedal Thomsen: Course root, Networks and Security



Finn Schirmer Andersen: Computer Architecture



David Gray Marchant: C programming, Operating Systems and Network programming

TAs:

- Tobias Andersen
- Lasses Helmer Larsen
- Markus Bjørn Madsen
- Sofus Ravnkilde Bloch Jensen

TAs will gladly help with

- Group members
- The right way to the administration
- A fellow student that can answer questions (or help find the answers)

Who to contact

- Your TA - Should be able to at least guide you in the right direction about anything
- Me/Finn - We are responsible for most of the course material and assignments
- Michael - Course admin and exam qualification

Discord is preferred over Absalon/email if its about a course topic. If you've got a question someone else probably does too.

Overall outline

Week 36-37 Data representation and machine model

Week 38 C programming

Week 39-40 Memory and operating systems

Week 41 Concurrent and parallel programming

Week 42 Fall break

Week 43-45 Computer networks - application and transport layer

Week 46 No activities (reexam week)

Week 47 Computer networks - security and efficiency

Week 48-51 Machine architecture

Week 52 Christmas vacation

Week 1-2 Computer networks and security - network and link layer

Week 4 4-hour written exam

Reading Material

COD Computer Organization and Design (RISC-V Edition), David A. Patterson and John L. Hennessy, second edition, ISBN: 978-0-12-820331-6

KR Computer Networking: A Top-Down Approach, James F. Kurose and Keith W. Ross, Pearson, 8th and Global Edition, ISBN 13: 978-1-292-40546-0 (This book will not be used before December)–7th edition is also acceptable

JG Modern C, Jens Gustedt,
<https://hal.inria.fr/hal-02383654/document>

OSTEP Operating Systems: Three Easy Pieces, Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau,
<https://pages.cs.wisc.edu/~remzi/OSTEP/>

?? Some notes and book chapters that will be made available through the detailed course schedule

COD is (and KR will be) available at Academic Books at Panum (<http://www.academicbooks.dk/>) and Polyteknisk Boghandel at Biocenteret (<http://www.polyteknisk.dk/>).

How to Use This Course

- Reading + Videos
 - ▶ Designed to compliment each other
 - ▶ Videos released the week before
 - ▶ Introduces the topics of the week
- Exercise classes
 - ▶ Will assume you've done some reading / watched the videos
 - ▶ First shot at playing around with the topics in practice
- Lectures
 - ▶ Go over material as more traditional lecture
 - ▶ Additional material not covered by videos
 - ▶ An oppourtunity to ask questions to me directly
- Cafes
 - ▶ Time for dedicated help with assignments

Both lecture material and exercises are important!

Exercises

- Mondays 15:15-17:00
 - Wednesdays 15:15-17:00
-
- Traditional format for explaining the topics
 - Ask questions!
 - I'll try and remember to release slides beforehand
 - Introduces most of the exam material

Exercises and Assignment Cafés

Exercises

- Mondays 13:15-15:00
- Wednesdays 13:15-15:00

Practice a lot of techniques used in assignments

Cafés

- Wednesdays 10:15-12:00

You can work on anything during either, but TAs are asked to prioritise exercises in exercise classes and assignments in cafes.

Details: <https://github.com/diku-compSys/compSys-e2025>. Also on Discord. See Absalon/Modules.

Assignments

- There are 5 assignment in total during the course with deadlines every two or three weeks (all Sundays). The assignments will be evaluated with points.
- Assignments will be awarded varying points (typically 6).
- You are required to achieve at least 50 % of the total number of points (equal to 13).
- Also we will require that you achieve points in each the of topics of the course to ensure that you have touched all parts of the curriculum.
- Assignments are made to be solved in groups of 2-3 students, but you *can* also do them alone.
- If you are resitting the course you do not need to resubmit assignments you've already completed, but practice is always best

Assignment rules

The Fundamental Principle of Group Assignments

Each group *must* make and justify their own solution.

This means

- You can talk with other people about the assignments: Teachers, TAs, other students, etc.
- You cannot share written code with other groups.
- You are not allowed to use code that you did not write yourself without proper citation.
- You cannot share written text with other groups.
- You are not allowed to use text of material without proper citation
 - ▶ This also includes material provided on the course.
- You are only allowed to use ChatGPT and other AI as you would any other person.
- *MUST* include a KU Generative AI Declaration
- Anything handed in you must be able to justify.

Assignments vs. exercises

- Note! Both are equally important
- Assignments:
 - ▶ Seek to test learning goals that relates to implementation and development of computer systems.
 - ▶ Do not fully prepare you for the written exam.
- Exercises:
 - ▶ Help you understand the theoretical parts of the material.
 - ▶ Prepare you for part of the exam.

Size

2-3 student advised. 1 can be accepted but not recommended. More than 3 is only allowed in special circumstances

- Sign up for classes with your group-mates on Absalon
- If you need one or more members
 - ▶ Look on announcements for details (use Absalon and Discord)
 - ▶ TA can help
- You can change groups at any time, but be reasonable to your colleagues

Tools

- RARS - RISC-V simulator
- C compiler – gcc (clang on macOS)
- C debugger – gdb (lldb on macOS)
- Profiler – Valgrind (perhaps leaks on macOS)
- You can also install all tools on you laptop
 - ▶ Linux: most available though apt
 - ▶ macOS: most available though Homebrew
 - ▶ Windows: Windows Subsystem for Linux
- Set up your tool chain
 - ▶ recommended using git to share code and reports in your group
 - ▶ Sign-up at GitHub today and apply for the *Student Developer Pack*
 - ▶ <https://education.github.com/>
- Tool-site is available on GitHub

- You are allowed to use them
- But we'd prefer you didn't for the programming tasks
- I won't be using them for this course, either to make or mark material
- Most of the programming tasks we're going to ask of you are not things you'd have to do in your regular work
- But the point is to do them to really learn what is important and why
- Learning these concepts are what makes you a Computer Scientist
- Remember the KU Generative AI Declaration

- September 2nd, 10.15-12.00
- DIKU 1-0-10, 1-0-18
- Introductions to many of the ancillary tools that will make life easier
 - ▶ Git
 - ▶ Unix
 - ▶ GDB
- Has been very helpful in previous years...

Exam

- A 4-hour written exam; Jan 27th 2026.
- The exam will be a BYOD-exam.
- The course syllabus is the exercises, assignments and reading material/videos.
- Previous exams will be made available nearer the time.

Registers

- I've been researching different methodologies for teaching and their effects on final grades
- But to present my findings I need actual data
- So I'll be collecting registers are lectures and exercise classes
- For research only! Will not affect your grades or whatnot, in fact probably won't be looked at till after the course is done
- Any research will only be in aggregate, e.g. everyone attending week 4 got a 12 on the exam
- Please be honest.
- KU numbers are most helpful, but anonymous responses are more helpful than nothing

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Everything is bits

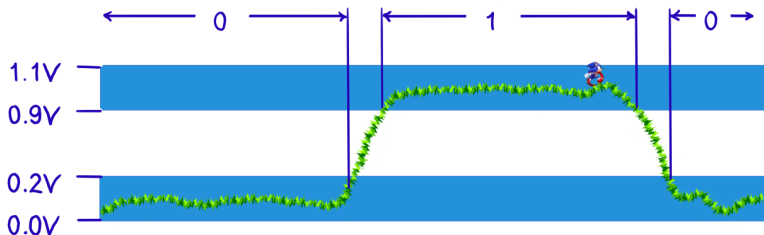
- Each bit is 0 or 1.
- By interpreting sets of bits in various ways...
 - ▶ ...computers determine what to do.
 - ▶ ...represent and manipulate numbers, sets, strings—*data*.

Why bits? Why not decimals? Could it have been some other way?

Everything is bits

- **Why bits? Electronic implementation.**

- ▶ Easy to store with bistable elements.
- ▶ Reliably transmitted on noisy and inaccurate wires (error correction).



- **... But there exist models that do not use bits.**

- ▶ The Soviet Setun computer used ternary *trits*.
- ▶ Quantum computers use *qubits* that are in a superposition of the two states.
 - ▶ ...error correction is the main challenge here.

Binary numbers

- **Base 2 number representation.**

- ▶ Represent 15213_{10} as 11101101101101_2
- ▶ Represent 1.20_{10} as $1.0011001100110011[0011] \dots_2$
- ▶ Represent 1.5213×10^4 as $1.1101101101101_2 \times 2^{13}$

- **Machine numbers are of some finite size.**

- ▶ If we use k bits to represent a number, only 2^k distinct values are possible.
- ▶ How we interpret those bits can vary.
- ▶ **Why do we use finite-sized numbers?**

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- ▶ How we interpret those bits can vary.
- ▶ **Why do we use finite-sized numbers?**
- ▶ A “ k -bit machine” handles numbers of up to k bits “natively” (meaning fast).

Encoding byte values

Byte = 8 bits

- (Machine-specific, but is true for all mainstream machines.)
- 256 different values.
- Binary 00000000_2 to 11111111_2 .
- Decimal 0_{10} to 255_{10} .
- Hexadecimal 00_{16} to FF_{16} .
 - ▶ Base 16 number representation.
 - ▶ Uses characters 0–9 and A–F.
 - ▶ In C we write $FA1D37B_{16}$ as
 - ▶ $0xFA1D37B$
 - ▶ $0xfa1d37b$ (case does not matter)

Hex	Dec	Bin
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

Example sizes of C types on various computers

C data type	Typical 16-bit	Typical 32-bit	Typical 64-bit	x86-64
char	1	1	1	1
short	1	2	2	2
int	2	4	4	4
long	4	4	8	8
int32_t	4	4	4	4
int64_t	8	8	8	8
float	4	4	4	4
double	8	8	8	8
long double	-	-	-	10
pointer	2	4	8	8

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Boolean algebra

Developed by George Boole in 19th century

- Algebraic representation of logic (“truth values”).
- Encode *true* as 1 and *false* as 0.

And	&	0	1
	0	0	0
	1	0	1

Or		0	1
	0	0	1
	1	1	1

Not	~	
	0	1
	1	0

Exclusive-or	^	0	1
	0	0	1
	1	1	0

- These operations can be implemented with tiny electronic *gates*.

General boolean algebras

- The truth tables generalise to operate on *bit vectors*, applied elementwise.

01101001	01101001	01101001	
& 01010101	01010101	^ 01010101	~ 01101001
01000001	01111101	00111100	10010110

- This is the form they take in programming languages such as C.

Bit-level operations in most C-like languages

Operations `&`, `|`, `~`, `^` available in C.

- Apply to any integral type.
 - ▶ E.g. `long`, `int`, `short`, `char`...
- Interpret operands as bit vectors.
- Applied bit-wise.

Examples

- $\sim 0x41 = 0xBE$
 - ▶ $\sim 01000001_2 = 10111110_2$
- $\sim 0x00 = 0xFF$
 - ▶ $\sim 00000000_2 = 11111111_2$
- $0x69 \ \& \ 0x55 = 0x41$
 - ▶ $01101001_2 \ \& \ 01010101_2 = 01000001_2$
- $0x69 \ \& \ 0x55 = 0x7D$
 - ▶ $01101001_2 \ \& \ 01010101_2 = 01111101_2$

Shift operations

■ Left shift $x \ll y$

- ▶ Shift bit-vector x left by y positions.
 - ▶ Throws away excess bits on the left.
 - ▶ Fills with zeroes on right.

■ Right shift $x \gg y$

- ▶ Shift bit-vector x right by y positions.
 - ▶ Throws away excess bits on the left.
- ▶ Logical shift: Fill with 0s on left.
- ▶ Arithmetic shift: Replicate most significant bit on left.

■ Undefined behaviour

- ▶ Shifting a negative amount or by the vector size or more.

x		01100010
$x \ll 3$		00010000
$x \gg 2$	(log)	00011000
$x \gg 2$	(arith)	00011000
x		10100010
$x \ll 3$		00010000
$x \gg 2$	(log)	00101000
$x \gg 2$	(arith)	11101000

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Encoding integers

Suppose x_i is the i th bit of a w -bit word (with x_0 being the least significant bit).

Unsigned

$$B2U(X) = \sum_{i=0}^{w-1} x_i \cdot 2^i$$

Two's complement

$$B2S(X) = -x_{w-1} \cdot 2^{w-1} + \sum_{i=0}^{w-2} x_i \cdot 2^i$$

```
int16_t x = 15213;  
int16_t y = -15213;
```

	Decimal	Hex	Binary
x	15213	3 B 5 D	0011 1011 0110 1101
y	-15213	C 4 9 3	1100 0100 1001 0011

Sign bit

- For 2's complement, most significant bit (x_{w-1}) indicates sign.
 - ▶ 0 for non-negative.
 - ▶ 1 for negative.

Two's complement encoding example

```
int16_t x = 15213; // 0011 1011 0110 1101
int16_t y = -15213; // 1100 0100 1001 0011
```

Weight	15213		-15213	
1	1	1	1	1
2	0	0	1	2
4	1	4	0	0
8	1	8	0	0
16	0	0	1	16
32	1	32	0	0
64	1	64	0	0
128	0	0	1	128
256	1	256	0	0
512	1	512	0	0
1024	0	0	1	1024
2048	1	2047	0	0
4096	1	4096	0	0
8192	1	8192	0	0
16384	0	0	1	16384
-32768	0	0	1	-32768
Sum	15213		-15213	

Let's play a game

`http://topps.diku.dk/compsys/integers.html`

Numeric ranges, here for 16-bit signed and unsigned integers

Unsigned

$$\text{UMin} = 0 = 0 \dots 0_2$$

$$\text{UMax} = 2^w - 1 = 1 \dots 1_2$$

Two's complement signed

$$\text{SMin} = -2^{w-1} = 10 \dots 0_2$$

$$\begin{aligned} \text{SMax} &= 2^{w-1} - 1 = 01 \dots 1_2 \\ &= -1 \qquad \qquad \qquad = 1 \dots 1_2 \end{aligned}$$

Values for $w = 16$:

	Decimal	Hex	Binary
UMax	65535	F F F F	1111 1111 1111 1111
SMax	32767	7 F F F	0111 1111 1111 1111
SMin	-32768	8 0 0 0	1000 0000 0000 0000
-1	-1	F F F F	1111 1111 1111 1111
0	0	0 0 0 0	0000 0000 0000 0000

Values for different word sizes

	W			
	8	16	32	64
UMax	255	65,535	4,294,967,295	18,446,744,073,709,551,615
SMax	127	32,767	2,147,483,647	9,223,372,036,854,775,807
SMin	-128	-32,768	-2,147,483,648	-9,223,372,036,854,775,808

Observations

$$|\text{SMin}| = \text{SMax} + 1$$

$$|\text{UMax}| = 2 \cdot \text{SMax} + 1$$

Note the assymetric range.

C Programming

- `#include <limits.h>`
- Declares constants, e.g:
 - ▶ `ULONG_MAX`
 - ▶ `LONG_MAX`
 - ▶ `LONG_MIN`
- Values are platform-specific.

Unsigned and signed numeric values (here $w = 4$)

x	$B2U(x)$	$B2S(x)$
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	-8
1001	9	-7
1010	10	-6
1011	11	-5
1100	12	-4
1101	13	-3
1110	14	-2
1111	15	-1

- **Equivalence**

- ▶ Same encoding for non-negative values.

- **Uniqueness**

- ▶ Every bit pattern represents distinct integer value.
- ▶ Each representable integer has unique bit encoding.
- ▶ The representation is **bijective**.

- **Can invert mappings**

- ▶ $U2B(x) = B2U^{-1}(x)$
 - ▶ Bit pattern for unsigned integer.
- ▶ $S2B(x) = B2S^{-1}(x)$
 - ▶ Bit pattern for two's complement integer.

Main takeaways

- Distinguish between **representation** and **interpretation**.
- Low-level values **do not describe their own structure**.
- Everything is built in layers.
- A good computer scientist adds new, clean, layers of abstraction.
 - ▶ A bad one adds layers that hide without simplifying.
 - ▶ A terrible one adds layers that complicate and obfuscate.
- **The point of this course is to show that there is no magic, only the work of careful people who put in a lot of effort.**